A discrete $\leftrightarrow$ continuous bridge for the Riemann zeta and Dirichlet eta functions: a zero-birth knob, a midpoint warp onto $\zeta(s, \frac{1}{2})$, and a unified $O(1/K)$ rate law

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Abstract

We study a one-parameter family of analytic objects that interpolates from the bland continuous integral $\int_1^\infty x^{-s}\,\mathrm{d}x = 1/(s-1)$ (which keeps ζ's pole but has *no* non-trivial zeros, no functional equation, and no arithmetic) to the full Dirichlet series $\zeta(s)$. The interpolation knob is the number K of Fourier harmonics of the discreteness kernel (the Euler–Maclaurin sawtooth, equivalently the Abel–Plana comb) that one retains. As K grows, the non-trivial zeros are *born* out of a literally zeroless endpoint and then *migrate* onto their critical lines. We exhibit two routes to the same destination, an *additive* comb that adds harmonics to the integrand and a *nonlinear midpoint warp* of the integration variable that is deliberately engineered to land on the half-shifted Hurwitz zeta $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$; and we record the general-phase warp onto $\zeta(s, \alpha)$, which reproduces the Saias–Weingartner $P \cdot L$ dichotomy (clean vertical zero strings only at $\alpha \in \{0, \frac{1}{2}, 1\}$). The unifying result is a single $O(1/K)$ *rate / birth-K law*: every route truncates the same sawtooth, so every route converges at the same algebraic rate, with a route-dependent displacement constant whose real part fixes the approach side and whose imaginary part is numerically tied to Gram's law. Finally we read the Dirichlet-η zeros as a two-component "crystal $\times$ GUE" spectrum and isolate the single prime $p = 2$. This is a *computational / experimental-mathematics* synthesis: every individual endpoint is classical and is cited as such; the contribution is the unifying construction, the rate law, and the spectral readings, not a new theorem about where zeros live. All figures and numbers are reproducible from the accompanying repository.

Contents

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1 Introduction

The Riemann zeta function $\zeta(s) = \sum_{n\geq 1} n^{-s}$ is a *discrete* object: a sum over the integers. Its simplest continuous shadow is the integral that replaces the sum by its leading Euler–Maclaurin term,

$$\int_1^\infty x^{-s}\,\mathrm{d}x = \frac{1}{s-1}, \qquad \operatorname{Re} s > 1, \tag{1}$$

which retains ζ's pole at $s = 1$ but is otherwise inert: it has *no* non-trivial zeros, no functional equation, and no arithmetic content whatsoever. The gap between (1) and the genuine $\zeta(s)$ is the discreteness of the integers, and the Euler–Maclaurin formula makes that gap explicit as a sum of correction terms built from the periodic Bernoulli "sawtooth" $\widetilde{B}_1(x) = \{x\} - \frac{1}{2}$. The Fourier series of the sawtooth,

$$\widetilde{B}_1(x) = \{x\} - \tfrac{1}{2} = -\sum_{k\geq 1} \frac{\sin(2\pi k x)}{\pi k}, \tag{2}$$

suggests an obvious interpolation knob: keep only the first K harmonics. At $K = 0$ one is back at the bland endpoint (1); as $K \to \infty$ one restores the full discreteness and recovers $\zeta(s)$.

This paper studies what the non-trivial zeros do along that interpolation. The short answer, made literal and visual throughout, is that the zeros are *born* as K leaves 0 and then *migrate* onto their critical lines as $K \to \infty$. Because the $K = 0$ endpoint is genuinely zeroless, this is a *creation-and-migration* mechanism, not a rearrangement of a pre-existing zero set; we contrast it carefully with the de Bruijn–Newman heat flow in Section 8.

Honest framing. We state at the outset, as a deliberate decision, that this is a *methods / experimental-mathematics* paper: a unified, reproducible re-derivation that threads several *known* endpoints onto one explicit "restore discreteness" knob and measures how the zeros respond. It is *not* a theorem about where the zeros of ζ live, and it carries no implication toward the Riemann hypothesis. Every individual ingredient below is classical, and we have made two adversarial passes through the literature to keep the attribution honest. What is classical, and what we believe is the surviving synthesis, is stated plainly here and revisited in Section 8:

- **Classical (we cite, we do not claim):** the half-shift identity $(2^s - 1)\zeta(s) = \zeta(s, \frac{1}{2})$ and the migration of Hurwitz-type zeros onto $\sigma = \frac{1}{2}$ [9, 13]; the privileged $\alpha \in \{0, \frac{1}{2}, 1\}$ $P \cdot L$ dichotomy [26], with generic off-line $\zeta(s, \alpha)$ zeros [5, 7]; the $\sigma = 1$ eta comb at $t = 2\pi k / \ln 2$ [2, 28]; and the appearance of prime-power log-frequencies (smallest $\ln 2$) in the zeta-zero number variance, in a regime where the GUE prediction fails [3, 16].
- **The synthesis we offer:** the harmonic-count K as a *zero-birth knob* attached to a literally zeroless endpoint; the Euler–Maclaurin sawtooth $\equiv$ Abel–Plana comb identity read as a zero-birth event; the midpoint warp *deliberately* landing on $\zeta(s, \frac{1}{2})$ (with its $\sigma = 0$ companion as the functional-equation mirror of η's $\sigma = 1$ comb); a *unified* $O(1/K)$ *rate law* tying the additive and warp routes together, with a Gram's-law coupling on the vertical axis; and a comb$\,\uplus\,$GUE two-component reading of the η zeros that isolates the single prime $p = 2$.

Roadmap. Section 2 sets up the two endpoints: the bland $1/(s-1)$ and the structured continuous-η integral, whose off-critical zero string already obeys the Riemann–von Mangoldt counting law. Section 3 restores discreteness *additively* and exhibits the zero-migration map. Section 4 restores it by a nonlinear *warp* of the integration variable, deliberately targeting $\zeta(s, \frac{1}{2})$, and leads with the algebra that makes the half-integer object *contain* ζ; a short supplement (Section 4.5) warps the η integrand itself and finds the midpoint phase annihilated by its cos sign-carrier. Section 5 generalizes the warp to arbitrary phase and recovers the $P \cdot L$ dichotomy. Section 6 is the unifying $O(1/K)$ rate law, with a stability corollary (Section 6.1): every finite-K interpolant is meromorphic on all of $\mathbb{C}$, with no abscissa of convergence. Section 7 reads the η zeros as a two-component spectrum. Section 8 foregrounds the prior-art deflators and gives the de Bruijn–Newman contrast, and Section 9 points to the reproducible code.

2 The two endpoints

2.1 The bland endpoint

For $\operatorname{Re} s > 1$, (1) is the entire continuous content of the leading Euler–Maclaurin term applied to ζ. Analytically continued, $1/(s-1)$ is a meromorphic function with a single simple pole and *no zeros at all.* It is the honest "$K = 0$" object on the ζ side: pole present, everything else absent.

2.2 The structured continuous endpoint

The alternating (Dirichlet) eta function $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s} = (1 - 2^{1-s})\zeta(s)$ has a more interesting continuous shadow, because the alternating sign is $(-1)^n = \cos(\pi n)$ at the integers. Replacing the sum by the corresponding integral gives

$$F(s) \;=\; \int_1^\infty \cos(\pi x)\, x^{-s}\, \mathrm{d}x, \tag{3}$$

which, unlike (1), is genuinely structured: it has a closed form in incomplete gamma functions and a non-trivial zero string. Numerically (see Fig. 1) the zeros of (3) obey three laws, confirmed to 3–5 digits over $8 < t < 160$:

$$\text{(real part)} \quad \sigma(t) \to \tfrac{3}{2}, \tag{4}$$

$$\text{(spacing)} \quad \Delta t \sim \frac{2\pi}{\ln(t/\pi)}, \tag{5}$$

$$\text{(counting)} \quad N(t) = \frac{t}{2\pi} \ln \frac{t}{\pi \mathrm{e}} + \frac{5}{8}, \tag{6}$$

the last of which is the Riemann–von Mangoldt counting form familiar from ζ itself [23, 30].

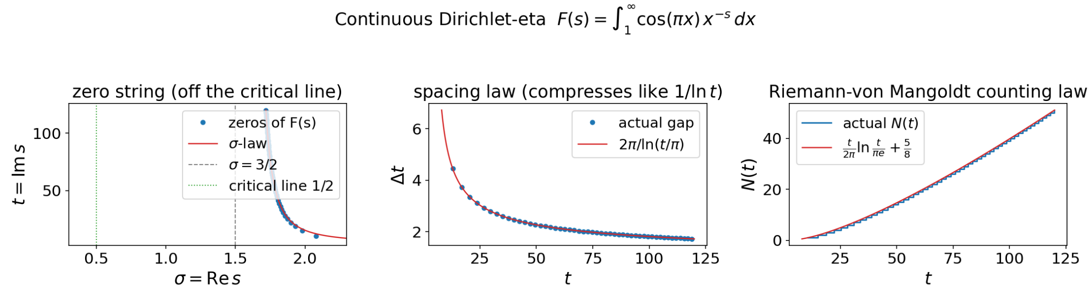

Figure 1: The continuous-η endpoint $F(s) = \int_1^\infty \cos(\pi x)x^{-s}\mathrm{d}x$ and its off-critical zero string. The zeros have the *right density* (the Riemann–von Mangoldt law (6)), but their real parts drift toward $\sigma = \frac{3}{2}$ (4): the continuum supplies "non-trivial zeros with the correct counting law" but not "zeros on the critical line."

Observation 1. The continuous endpoint already supplies non-trivial zeros with the *correct Riemann–von Mangoldt density* (6); what it withholds is their *location*. The zeros live off the critical line, drifting toward $\sigma = \frac{3}{2}$. "Zeros with the right counting law" is therefore a property of the continuum; "those zeros on $\mathrm{Re}\, s = \frac{1}{2}$" is what restoring discreteness must supply. Closing that gap is the whole point of the bridge.

3 Phase 1: the additive comb

The first route restores discreteness *additively*: add the Fourier harmonics (2) of the discreteness kernel back into the integrand, one at a time, and track each zero as a curve in K. We call the resulting trajectories the *zero-migration map* (Fig. 2).

Two integral guises of one comb. The discreteness correction admits two classical integral representations that we verify agree term by term: the Euler–Maclaurin sawtooth $\widetilde{B}_1(x) = \{x\} - \frac{1}{2}$ and the Abel–Plana "Bose" comb $1/(\mathrm{e}^{2\pi x} - 1)$. They are the same object viewed through two windows; truncating either at K harmonics gives the identical interpolant. We read this shared truncation as the engine of the rate law in Section 6.

Birth on the ζ side. On the ζ side the zeroless endpoint is $1/(s-1) + \frac{1}{2}$ (the integral plus the constant Euler–Maclaurin term). As the *first* harmonic switches on, the non-trivial zeros are literally *born*, and they then slide onto $\sigma = \frac{1}{2}$ as K increases. There is no zero to "move" at $K = 0$; the zero set is created.

Splitting on the η side. On the η side the $K = 0$ ground string *splits* as K grows: half the zeros onto $\sigma = \frac{1}{2}$ (the genuine ζ zeros) and half onto $\sigma = 1$ (the zeros of the prefactor $1 - 2^{1-s}$, located at $t = 2\pi m/\ln 2$). This split is a near-bijection of densities: writing the two pieces out,

$$\underbrace{\frac{t}{2\pi}\ln\frac{t}{2\pi\mathrm{e}}}_{\sigma=1/2\ \text{(critical line)}} + \underbrace{\frac{t\ln 2}{2\pi}}_{\sigma=1\ \text{(prefactor comb)}} = \frac{t}{2\pi}\ln\frac{t}{\pi\mathrm{e}}, \tag{7}$$

which recovers the continuous endpoint's density (6). No zeros are created or destroyed in the η migration; the existing string is *sorted* onto two lines, and the driver's counting check witnesses (7) numerically.

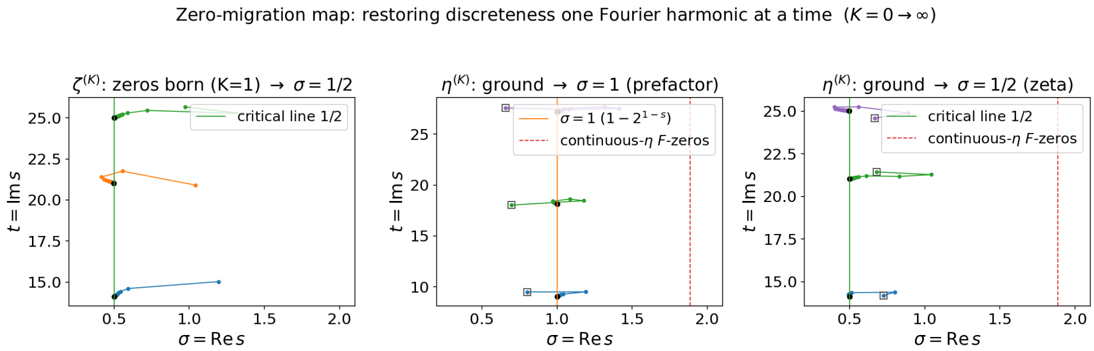

Figure 2: The additive zero-migration map: restoring discreteness one harmonic at a time. On the ζ side the zeros are *born* as the first harmonic switches on and slide onto $\sigma = \frac{1}{2}$; on the η side the ground string *splits* onto $\sigma = \frac{1}{2}$ (the ζ zeros) and $\sigma = 1$ (the $1 - 2^{1-s}$ comb at $t = 2\pi m/\ln 2$), conserving the density (7).

4 Phase 2: the nonlinear midpoint warp

The second route reaches the critical line by a different mechanism: instead of adding harmonics to the integrand, we *warp the integration variable* so the measure is pulled toward integer plateaus, $n^* = x + \varphi_K(x)$, where φ_K is the K-harmonic truncation of the sawtooth displacement. As $K \to \infty$ the warp collapses each unit cell onto its *midpoint*, so its natural target is the half-shifted Hurwitz zeta. Before the figures, we state the algebra that makes this target legitimate, which is the single most likely reader sticking point.

4.1 Why the half-integers reproduce ζ's own zeros

A fair worry: the literal Dirichlet series is summed over the counting numbers $1, 2, 3, \ldots$ (that is the $\alpha = 1$ warp of Section 5), so why should a *half-integer* object reproduce ζ's zeros and their migration at all? Because the half-integer object is not a different beast that merely mimics ζ; it *contains* ζ *as a factor*, by an exact identity. The half-integers are the odd integers rescaled by $\frac{1}{2}$:

$$\zeta(s, \tfrac{1}{2}) = \sum_{n \geq 0} (n + \tfrac{1}{2})^{-s} = 2^s \sum_{n \geq 0} (2n+1)^{-s} = 2^s \sum_{m \text{ odd}} m^{-s}, \tag{8}$$

and "sum over the odds" is "all integers minus the evens":

$$\sum_{m \text{ odd}} m^{-s} = \zeta(s) - 2^{-s}\zeta(s) = (1 - 2^{-s})\zeta(s), \tag{9}$$

so that

$$\boxed{\zeta(s, \tfrac{1}{2}) = 2^s(1 - 2^{-s})\zeta(s) = (2^s - 1)\,\zeta(s).} \tag{10}$$

This is *algebra, not a limit* (it is the Hurwitz half-shift; see 9, 13 and 8, §25.11). Summing over the half-integers equals summing over the counting numbers times the explicit entire factor $2^s - 1$. The zeros therefore simply split,

$$\{\text{zeros of } (2^s - 1)\zeta(s)\} \;=\; \underbrace{\{\zeta\text{'s zeros on } \sigma = \tfrac{1}{2}\}}_{\zeta\text{'s own, identically}} \cup \underbrace{\{\, s : 2^s = 1 \,\}}_{\sigma=0,\ t=2\pi k/\ln 2}, \tag{11}$$

since $2^s = e^{s \ln 2} = 1$ forces $\sigma = 0$ and $t = 2\pi k/\ln 2$. So the $\sigma = \frac{1}{2}$ family that the warp migrates onto *is* ζ's genuine non-trivial zero set, identically, and not a look-alike.

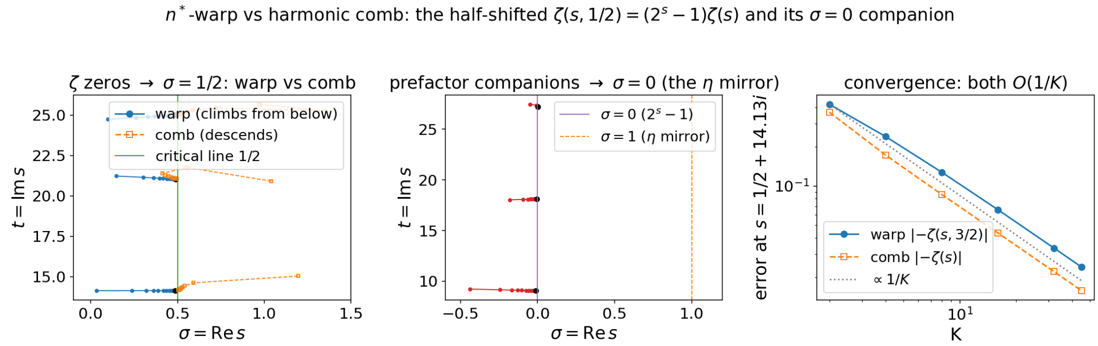

Figure 3: The nonlinear midpoint warp vs. the additive comb. The warp collapses each unit cell onto its midpoint, so it targets $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$ on the nose: ζ's zeros climb onto $\sigma = \frac{1}{2}$ *from below*, and the $2^s - 1$ companion migrates onto $\sigma = 0$ (the functional-equation mirror of η's $\sigma = 1$ comb). The two routes reach the critical line by *opposite* approaches, a contrast the rate law (Section 6) explains quantitatively.

Remark 1. The half-integer lattice carries the same prime content as ζ for every odd prime and isolates only $p = 2$ (the pulled-out Euler factor $1 - 2^{-s}$). That isolated factor reappears as the $2^s - 1$ companion on $\sigma = 0$, the functional-equation mirror of η's $\sigma = 1$ comb (Section 7) at the same heights. Using the counting numbers directly ($\alpha = 1$) would give plain $\zeta(s)$ and *only* the $\sigma = \frac{1}{2}$ family; the half-shift gives the *same* critical zeros plus that explicit, fully understood $\sigma = 0$ bonus, and does so from the pristine zeroless endpoint (1).

4.2 The warp and its zeros

With (10) in hand, the warp's behavior is clean (Fig. 3). The genuine ζ zeros climb onto $\sigma = \frac{1}{2}$ but, unlike the comb's zeros (born high, near $\sigma \approx 1.2$, and then descending), the warp's zeros are born *low* (near $\sigma \approx 0$) and climb *from below*. A companion family, the zeros of the $2^s - 1$ prefactor at $t = 2\pi k/\ln 2$, migrates onto $\sigma = 0$. Below $T = 40$ we measure a clean bijection: six zeros on $\sigma = \frac{1}{2}$ and four on $\sigma = 0$, matching the predicted counts from (11).

Observation 2 (Birth and migration are successive stages, not alternatives)**.** For the half-shift phase $\alpha = \frac{1}{2}$ the $K = 0$ warp object is the zeroless integral (1), $1/(s-1)$, with no non-trivial zeros at all, so at the start there is nothing to rearrange. The zeros are therefore *created* as harmonics are restored, and only afterwards do they move. This matches the integer phase $\alpha = 1$, which shares the same zeroless $K = 0$ endpoint; it is unlike the η side of Section 3, whose $K = 0$ ground string already carries zeros that then split. The lowest zeros appear already at the first harmonic, $K = 1$, and higher ones only as more harmonics are added (quantified by the birth-K law of Section 6). So for $\alpha = \frac{1}{2}$ both families are genuinely born out of the zeroless endpoint: the $\sigma = \frac{1}{2}$ family (the genuine ζ zeros of (11)) appears low, near $\sigma \approx 0$, and then migrates up onto $\sigma = \frac{1}{2}$, while the $\sigma = 0$ companions appear and settle onto $\sigma = 0$. The word "born" thus names the creation of a zero out of a zeroless endpoint, and "migrate" the subsequent motion onto a line; they describe two stages of one trajectory rather than a choice between them. That creation, not a shuffling of some fixed pre-existing set, is what sets the construction apart from the de Bruijn–Newman flow (Section 8).

4.3 The warp coordinate, visualized

Figure 4 pictures the warp itself rather than its zeros. The warp coordinate $n^*(x) = x + \varphi_K(x)$ starts as the linear x at $K = 0$ and, as harmonics are restored, bends into the midpoint staircase

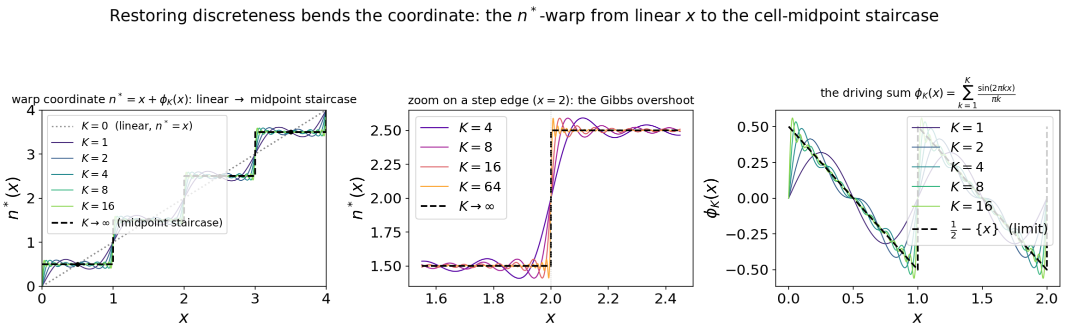

Figure 4: The warp coordinate $n^*(x) = x + \varphi_K(x)$ bending from the linear x ($K = 0$) into the midpoint staircase $\lfloor x \rfloor + \frac{1}{2}$ as Fourier harmonics are restored. Panel (b): the ~9% Gibbs overshoot at the step edges, which narrows but never vanishes, the source of the warp's non-elementary rate constant.

	steps land on	$K = 0$ endpoint	$K \to \infty$ target
$\alpha = \frac{1}{2}$ (midpoint, headline)	half-integers $\frac{1}{2}, \frac{3}{2}, \ldots$	x (pristine $1/(s-1)$)	$(2^s - 1)\zeta(s) = \zeta(s, \frac{1}{2})$
$\alpha = 1$ (integer)	counting numbers $1, 2, 3, \ldots$	$x + \frac{1}{2}$ (DC offset on)	$\zeta(s, 1) = \zeta(s)$

Table 1: The trade-off: a zero-mean periodic correction added to x can only ever reach $\lfloor x \rfloor + \frac{1}{2}$. Landing on the integers *requires* the non-trigonometric constant $\frac{1}{2}$, so at $K = 0$ one is already at $x + \frac{1}{2}$, no longer the bland trig-free x. One can have a pure-x start *or* integer steps, not both.

$\lfloor x \rfloor + \frac{1}{2}$, collapsing each unit cell onto its midpoint. The cell midpoints $m + \frac{1}{2}$ are exact fixed points at every K. Because the underlying sawtooth has a unit jump at each integer, the partial sums carry the classic ~9% Gibbs overshoot at the step edges, which narrows but never vanishes. That never-vanishing Gibbs boundary layer is precisely why the warp's rate constant is a non-elementary sine-integral rather than the comb's clean $s/2\pi^2$ (Section 6).

4.4 Half-integers vs. counting numbers: why $\alpha = \frac{1}{2}$ is the headline

Why does the staircase land on the half-integers rather than the counting numbers? Both are reachable: they are the $\alpha = \frac{1}{2}$ and $\alpha = 1$ cases of the general-phase warp (Section 5), whose cells collapse onto $\lfloor x \rfloor + \alpha$ via the DC-shifted displacement $\varphi_K^{(\alpha)}(x) = (\alpha - \frac{1}{2}) + \varphi_K(x) \to \alpha - \{x\}$. Figure 5 puts them side by side. The obstruction to having both is the constant ($k = 0$, DC) Fourier coefficient: the $\alpha = \frac{1}{2}$ displacement is the *pure* sawtooth φ_K (zero mean), while $\alpha = 1$ carries a constant $+\frac{1}{2}$.

This is why the bridge's headline is $\alpha = \frac{1}{2}$: it is the one phase that keeps the pristine zeroless endpoint (1) *and* lands on a clean line, and that line is the richer half-shifted $(2^s - 1)\zeta(s)$, whose $\sigma = \frac{1}{2}$ zeros are ζ's own (Section 4.1) and which also carries the $\sigma = 0$ companion. Choosing $\alpha = 1$ recovers plain $\zeta(s)$ and the literal counting numbers, but at the cost of the pristine endpoint *and* the companion line.

4.5 Warping the η integral: why $\alpha = \frac{1}{2}$ is a ζ-only luxury

The warp has so far acted only on the *bland* integrand x^{-s}. It is natural to ask what it does to the bridge's *structured* endpoint, the continuous-η integral (3): warp the whole integrand,

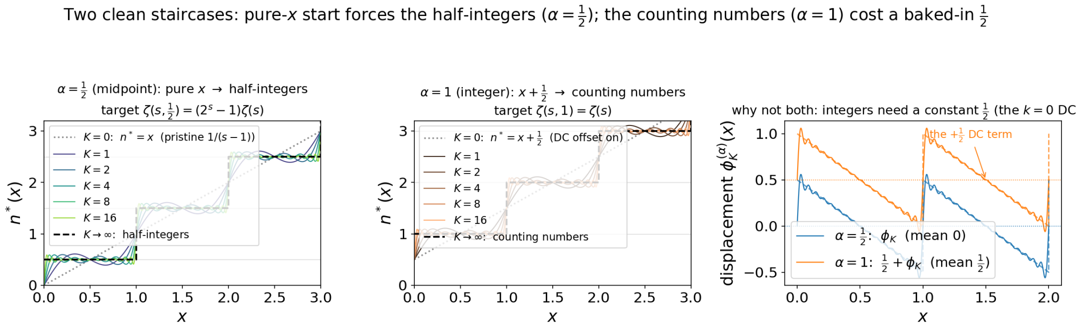

Figure 5: Two clean staircases: $\alpha = \frac{1}{2}$ (half-integers) and $\alpha = 1$ (counting numbers). Panel (c) shows the obstruction: the $\alpha = 1$ displacement carries a constant DC term while $\alpha = \frac{1}{2}$ does not. The half-integer midpoint is the *unique* clean staircase reachable from a pure-x start by pure trigonometric corrections, which is why it is the bridge's headline phase.

sign-carrier and all,

$$\int_1^\infty \cos(\pi n^*)\,(n^*)^{-s}\,dx, \qquad n^* = x + (\alpha - \tfrac{1}{2}) + \varphi_K(x). \tag{12}$$

The period-by-period collapse is $\sum_{m\geq1}(-1)^m \int_0^1 \cos(\pi g)(m+g)^{-s}\,du$ with $g(u) = u + (\alpha - \frac{1}{2}) + \varphi_K(u)$, and as $K \to \infty$ each cell lands on $\cos(\pi(m+\alpha))\,(m+\alpha)^{-s}$. The phase now matters in a way it did not for ζ:

- **The midpoint phase $\alpha = \frac{1}{2}$ is annihilated.** $\cos(\pi(m+\frac{1}{2})) = 0$ identically — the midpoint lattice $\{m + \frac{1}{2}\}$ is *exactly the zero set of the η sign-carrier* $\cos(\pi x)$ — so (12) collapses to 0 (numerically as $O(1/K)$, the finite-K residual being the Gibbs smoothing of those zeros). The elegant half-shift of Section 4 has *nothing to land on.*
- **Only the integer phase $\alpha = 1$ survives.** There $\cos(\pi(m + 1)) = (-1)^{m+1}$, so (12) $\to -\eta(s) + 1$; restoring the omitted $n = 1$ cell $\cos\pi \cdot 1^{-s} = -1$ (the load-bearing lower-endpoint term, the η-form of the warp's $+2^s$) completes it to $\eta(s) = (1-2^{1-s})\zeta(s)$. Its zeros are the genuine ζ zeros on $\sigma = \frac{1}{2}$ together with the $1-2^{1-s}$ comb on $\sigma = 1$ — the *same* split the additive comb produced in Section 3, now reached by the independent warp mechanism (Fig. 6), at the same $O(1/K)$ rate.

This sharpens Section 4.4: ζ could afford $\alpha = \frac{1}{2}$ — keeping the pristine zeroless endpoint *and* a clean line — precisely because its integrand x^{-s} has no zeros on the half-integer lattice. The η integrand does, so η is forced onto $\alpha = 1$ and pays the DC-offset endpoint that ζ avoided. The half-shift is special for ζ because the discreteness it samples carries no alternating sign.

5 The general phase and the $P \cdot L$ dichotomy

Generalizing the warp to pin cells to an arbitrary phase $n + \alpha$, $\alpha \in (0, 1)$, the completed target is the Hurwitz zeta $\zeta(s, \alpha)$. Sweeping α and watching where the zeros go exhibits the Saias–Weingartner $P \cdot L$ dichotomy [26]: clean vertical zero strings appear *only* at $\alpha \in \{0, \frac{1}{2}, 1\}$, the phases at which $\zeta(s, \alpha)$ factors as a polynomial in prime powers times a single Dirichlet L-function. (For $\alpha = \frac{1}{2}$ this is (10); in the language of 26 the relevant identity is $(1-2^{-s})\zeta(s) = P(s)\,L(s, \chi_0)$, with 2^s entire and nonvanishing.) Every *generic* α gives a Davenport–Heilbronn scattered set, with zeros off any vertical line and even into $\sigma > 1$ [5, 7, 26]; see also the value-distribution results for algebraic-irrational α in Sourmelidis and Steuding [29].

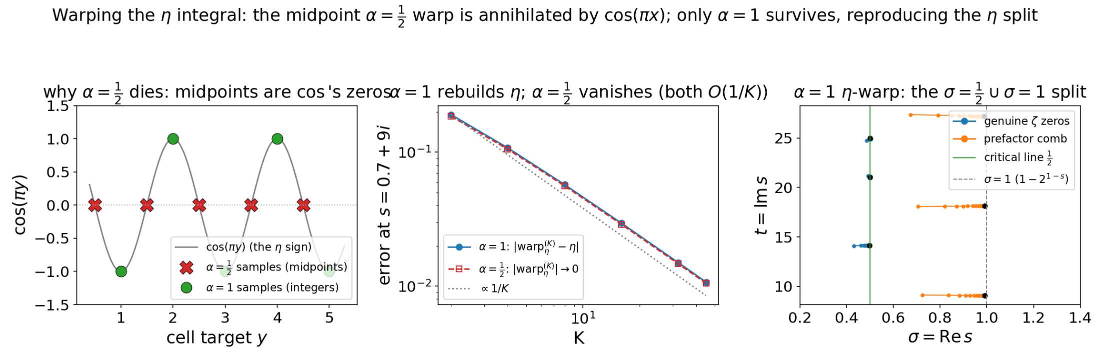

Figure 6: Warping the η integral (12). *Left:* the obstruction — the midpoints $m + \frac{1}{2}$ that the $\alpha = \frac{1}{2}$ warp samples are exactly the zeros of the η sign-carrier $\cos(\pi x)$, while the integers m that $\alpha = 1$ samples sit on its extrema ± 1. *Middle:* consequently $\alpha = 1$ rebuilds η at $O(1/K)$ while the $\alpha = \frac{1}{2}$ warp vanishes. *Right:* the $\alpha = 1$ η-warp's zeros migrate onto $\sigma = \frac{1}{2}$ (genuine ζ zeros) and $\sigma = 1$ (the prefactor comb) — the same split as the additive comb (Fig. 2), by a second, independent route.

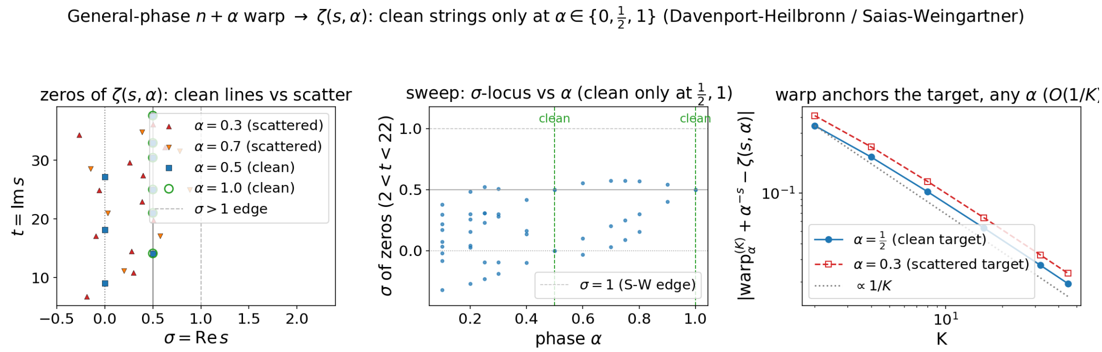

Figure 7: The general-phase $n + \alpha$ warp onto $\zeta(s, \alpha)$, exhibiting the Saias–Weingartner dichotomy: clean vertical strings only at $\alpha \in \{0, \frac{1}{2}, 1\}$; every generic α is Davenport–Heilbronn scattered off-line. This locates the half-shift of Section 4 inside a one-parameter family and explains why $\frac{1}{2}$ is *special* rather than merely chosen.

This locates the half-shift of Section 4 inside a one-parameter family and explains why $\frac{1}{2}$ is special rather than merely chosen: it is one of the three privileged phases. The "clean lines only at $\{0, \frac{1}{2}, 1\}$" picture is the load-bearing control that keeps the half-shift route honest.

6 The unified $O(1/K)$ rate law

The synthesis is that one mechanism sits behind every route. All routes truncate the *same* sawtooth (2), whose retained tail has power

$$\sum_{k>K} \frac{1}{(\pi k)^2} \sim \frac{1}{\pi^2 K}, \tag{13}$$

so every route converges as $O(1/K)$. A single zero-displacement law,

$$s_K - \rho \sim \frac{a_1(\rho)}{K}, \tag{14}$$

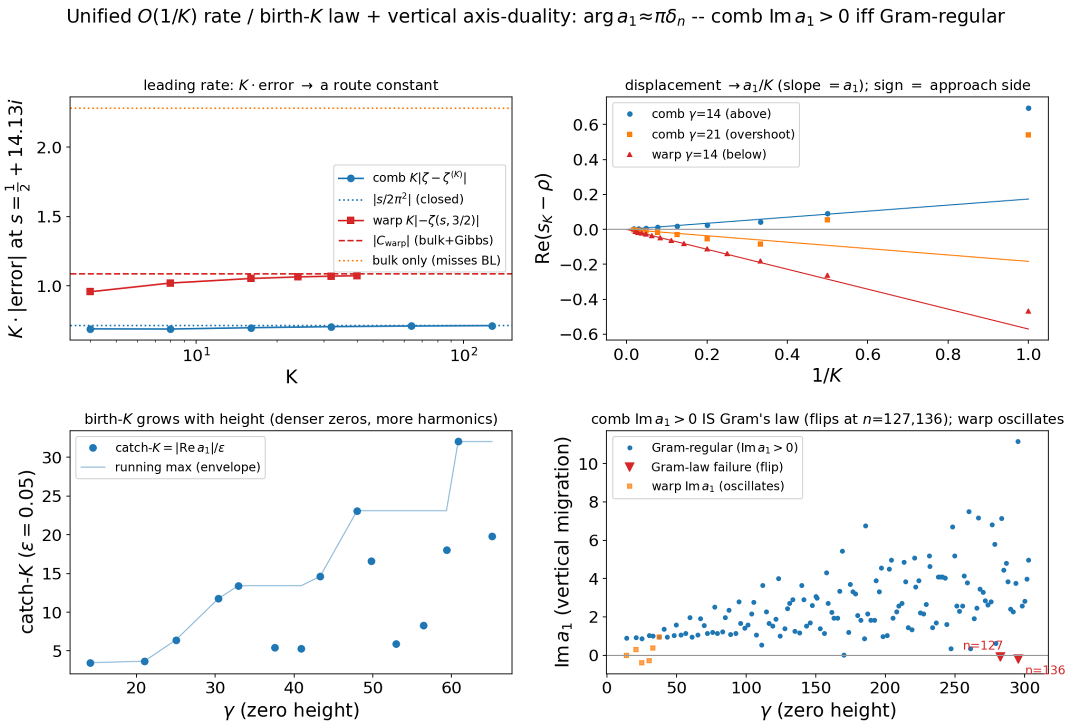

Figure 8: The unified $O(1/K)$ rate / birth-K law and the vertical axis-duality. All routes truncate the same sawtooth, so all converge as $O(1/K)$ (13); the displacement constant $a_1(\rho)$ of (14) controls approach side (sign $\mathrm{Re}\, a_1$), birth order ($K_{\mathrm{catch}} = |\mathrm{Re}\, a_1|/\varepsilon$), and the vertical migration ($\mathrm{Im}\, a_1$), whose sign tracks Gram's law.

covers all five zero families (the ζ comb, the η split onto $\sigma = \frac{1}{2}$ and $\sigma = 1$, and the warp's $\sigma = \frac{1}{2}$ and $\sigma = 0$ families), with a route- and zero-dependent complex displacement constant $a_1(\rho)$. The structure of a_1 carries the qualitative behavior:

- **Approach side** $= \mathrm{sign}(\mathrm{Re}\, a_1)$**.** The real part of a_1 fixes which side the zero approaches its line from, which is why the comb descends (and can overshoot) while the warp climbs from below and never overshoots; overshoot occurs precisely when $\mathrm{Re}\, a_1 < 0$.
- **Birth-K law.** The first K that resolves a given zero to tolerance ε is $K_{\mathrm{catch}} = |\mathrm{Re}\, a_1|/\varepsilon$. This ties the birth order of zeros to the Montgomery a-values $1/|\zeta'(\rho)|$.
- **Vertical migration = Gram's law.** The imaginary part $\mathrm{Im}\, a_1$ gives the vertical (along-t) migration. Its sign tracks Gram's law numerically: it first flips at the classical Gram-law failures (zeros $n = 127, 136, 196$).

This is where the package stops being several separate demonstrations and becomes one law: a single rate constant and displacement coefficient, route by route, plus a Gram's-law coupling on the vertical axis. It is the synthesis around which the whole study is organized.

6.1 Stability of the interpolants: no abscissa, height is the cost

The rate law also settles a question the Dirichlet series provokes: how far into the plane do the truncated objects remain valid? The series $\sum n^{-s}$ converges only for $\mathrm{Re}\, s > 1$ and its alternating cousin only for $\mathrm{Re}\, s > 0$, so one might expect $\zeta^{(K)}$ to inherit a barrier. It does not. Each $\zeta^{(K)}$ is *not* a partial sum of a Dirichlet series but $1/(s-1) + \frac{1}{2}$ plus finitely many harmonics, every one of which is an incomplete-gamma closed form — entire in s for its fixed nonzero argument.

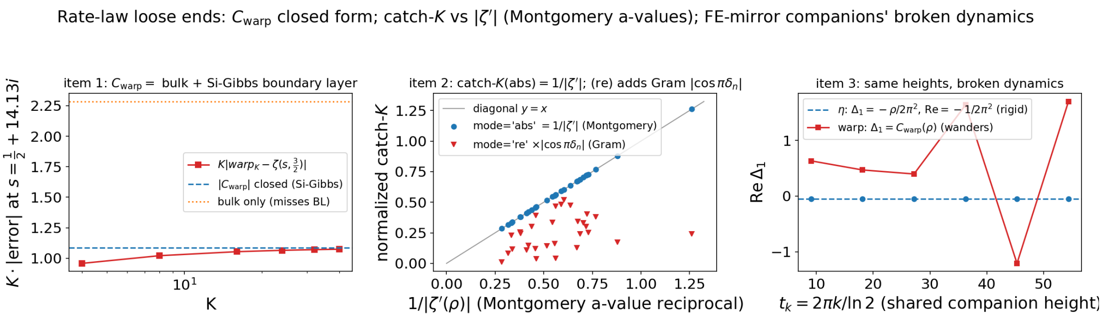

Figure 9: Three loose ends of the rate law: (i) a closed form for the warp constant C_{warp} as a Gibbs boundary-layer (sine-integral) integral; (ii) the tie of the birth threshold K_{catch} to the Montgomery a-values $1/|\zeta'(\rho)|$; and (iii) the fact that the $\sigma = 1/\sigma = 0$ companions share heights but *not* migration dynamics.

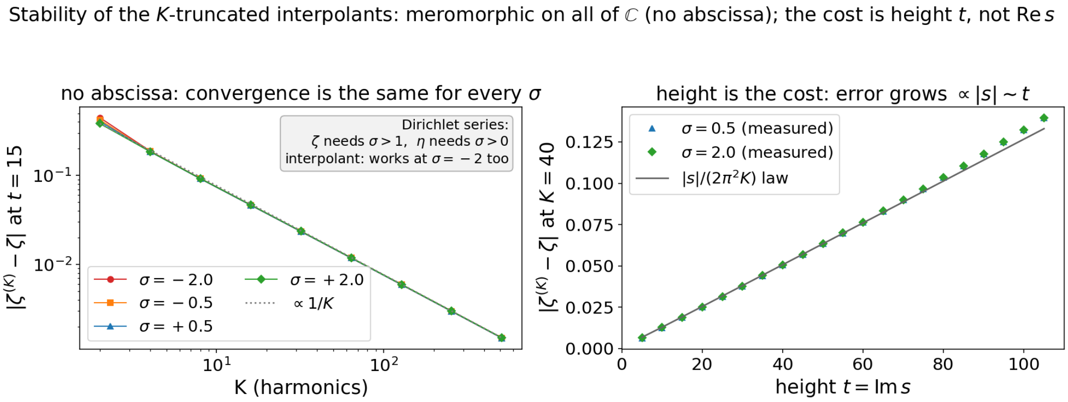

Figure 10: Where the K-truncated interpolants are stable. *Left:* the convergence error $|\zeta^{(K)} - \zeta|$ vs. K collapses onto one $O(1/K)$ line for every σ, including $\sigma < 0$ where both Dirichlet series diverge — there is no abscissa. *Right:* at fixed K the error grows linearly with height t, tracking the analytic rate constant $|s|/(2\pi^2 K)$: the cost is height, not Re s.

So every finite-K interpolant is meromorphic on all of $\mathbb{C}$, with the single pole at $s = 1$ and *no abscissa of convergence at all*: the zeroless endpoint $1/(s-1)$ already shows this, and each harmonic preserves it. The interpolant is its own analytic continuation, strictly better behaved than either series.

The cost of reaching deep into the plane is therefore not a half-plane boundary but *height*, and two facts make this precise (Fig. 10). First, Re s is essentially free: $\zeta^{(K)} \to \zeta$ at the *same* $O(1/K)$ rate at $\sigma = -2$ as at $\sigma = 2$, the convergence curves for $\sigma \in \{-2, -\frac{1}{2}, \frac{1}{2}, 2\}$ lying on top of one another. Second, the cost is set by $t = \text{Im}\, s$ through the comb rate constant $\Delta_1 = s/(2\pi^2)$ of Section 6 (the sawtooth tail (13)): the fixed-K error tracks $|s|/(2\pi^2 K)$, growing linearly with height. This is the convergence-side face of the birth-K law — higher, denser zeros need proportionally more harmonics — and it depends on t, not on any vertical line in σ.

The only genuine limit is finite precision, and it too is governed by height: at large t the moment evaluators carry $e^{\pi t/2}$-scale cancellations, so a fixed working precision eventually floors (we observe this near $|\,\text{Im}\, s| \approx 110$ and remove it by scaling precision, term count and quadrature degree with $|\,\text{Im}\, s|$). But the $O(1/K)$ convergence is slow enough that at moderate height this floor sits far below the achievable error and the descent stays clean; in exact arithmetic there is no divergence at any K — more harmonics always help.

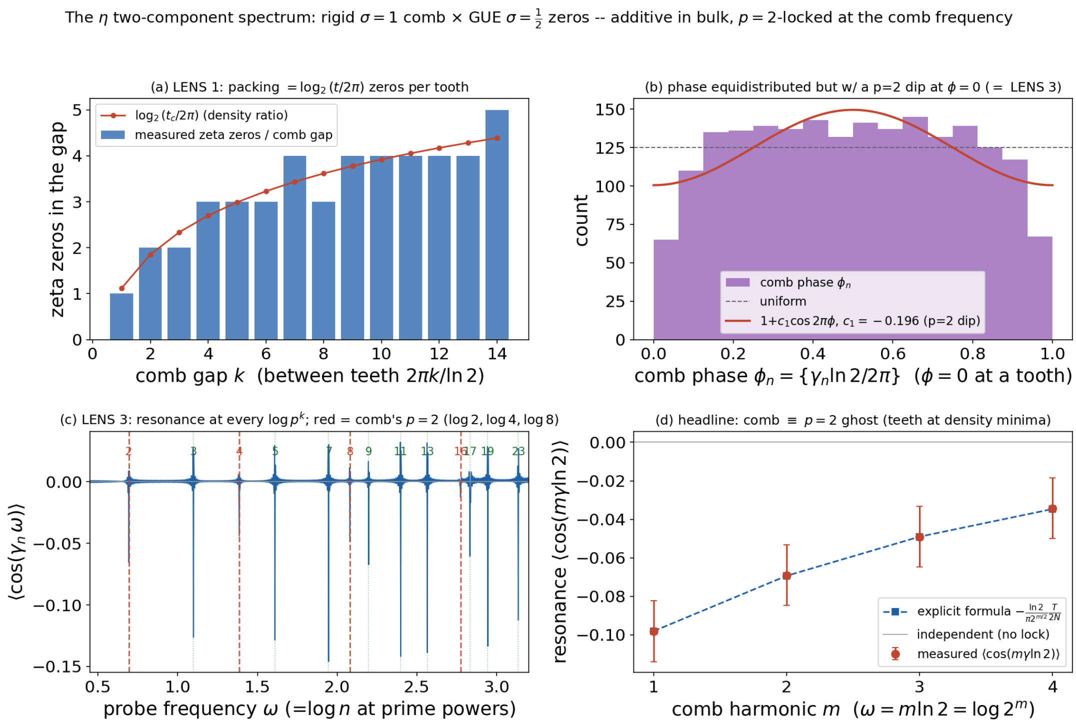

Figure 11: The η zeros as a two-component "crystal $\times$ GUE" spectrum: a rigid $\sigma = 1$ comb at $t = 2\pi k/\ln 2$ superposed on the $\sigma = \frac{1}{2}$ GUE ζ-zeros. The comb frequencies $m \ln 2$ are the $p = 2$ prime-power frequencies; probing the zeros there resonates with the explicit-formula coefficient (the "$p = 2$ ghost").

7 The two-component η spectrum

Projected onto the t-axis, the η zero set is a *rigid crystalline comb* ($\sigma = 1$ teeth at $t = 2\pi k/\ln 2$, perfectly periodic) *superposed* with a *GUE-random sequence* (the $\sigma = \frac{1}{2}$ ζ zeros) [18, 20]. Figure 11 applies three lenses:

1. **One-point packing.** There are $\log_2(t/2\pi)$ zeros per comb gap, equidistributed.
2. **Two-point number variance.** In the smooth bulk both lenses say the components behave as an *independent superposition*.
3. **Resonance at the comb frequency.** The third lens shows they are *not* independent. The comb teeth sit at the $p = 2$ prime-power frequencies $m \ln 2$, and by the explicit formula the ζ-zero density carries an oscillation at every prime-power frequency, so probing the zeros at the comb frequency *resonates*: the measured $\langle \cos(m\gamma \ln 2)\rangle$ matches the $p = 2^m$ explicit-formula coefficient to $\sim$4 digits, with a negative sign (the comb teeth sit at density *minima*).

The $\sigma = 1$ comb is, in this sense, the "$p = 2$ ghost" of the zeros' own prime structure.

Remark 2 (Honest caveat)**.** The appearance of $\ln 2$ in the zeros' number variance is *not* new. Berry [3], made rigorous by Lugar, Milinovich and Quesada-Herrera [16], already places prime-power log-frequencies in the zeta-zero number variance in a regime where the GUE prediction fails, and $\ln 2$ is the *single lowest* such frequency (note $\ln 3 < \ln 4 = 2\ln 2$, so $\ln 2$ is lowest but the band is not simply the multiples of $\ln 2$). What is ours here is only the comb $\uplus$ GUE *superposition* model, the *single-prime isolation* ($p = 2$), and the *eta tie-in* (relating the $\sigma = 1$ eta comb to that lattice), not the $\ln 2$ frequency itself. See Section 8.

8 Relation to prior work

The endpoints and most ingredients of this study are classical; we cite them rather than claim them. We have made two adversarial passes through the literature, and we foreground the two closest prior-art results explicitly so a reader can locate our residual contribution precisely.

8.1 Cite, do not claim

- The half-shift $(2^s - 1)\zeta(s) = \zeta(s, \frac{1}{2})$, its $\sigma = 0$ companion at $t = 2\pi k/\ln 2$, and the migration of Hurwitz-type zeros onto $\sigma = \frac{1}{2}$ as a parameter varies: Garunkštis and Steuding [9], Gonek [13]; see also Garunkštis and Tamošiūnas [10] for the periodic-Hurwitz Riemann–von Mangoldt counting law.
- The $\alpha \in \{0, \frac{1}{2}, 1\}$ "clean lines only" dichotomy: Saias and Weingartner [26]; generic off-line $\zeta(s, \alpha)$ zeros: Cassels [5], Davenport and Heilbronn [7].
- The $\sigma = 1$ eta comb at $t = 2\pi k/\ln 2$: Sondow [28]; tabulated by Beliakov and Matiyasevich [2]; 8, §25.2.3.
- The closest "knob" cousins use partial-sum *length* N rather than the harmonic count K, and none start from a zeroless endpoint: Gonek and Ledoan [11], Gonek and Montgomery [12]; see also Platt and Trudgian [21]. Zeros of combinations of Euler products for $\sigma > 1$: Booker and Thorne [4], Righetti [24]. A numerical study of the off-line zeros of the Davenport–Heilbronn counterexample (by a deform-and-track homotopy): Balanzario and Sánchez-Ortiz [1].

8.2 The deflator to foreground (the $p = 2$ ghost)

As stated in Remark 2, the strongest prior art for the two-component reading is Berry [3], proved by Lugar et al. [16]: the number variance $\Sigma^2(L)$ of the Riemann zeros has a prime-controlled regime in which the GUE prediction explicitly fails, governed by a von Mangoldt cosine sum at frequencies $\ln n$ whose single smallest is $\ln 2$. That work treats the *single* zeta-zero spectrum; it does not treat the Dirichlet-η / partial-sum zeros, does not build the comb $\uplus$ GUE two-spectrum superposition statistics, does not isolate a single prime, and does not tie the $\sigma = 1$ eta comb at $t = 2\pi k/\ln 2$ to the $p = 2$ lattice via the explicit formula. Those four are the surviving contribution of Section 7. Related "quasicrystal" readings of the zeros (the zero-spacing distributions of differenced L-functions [15], and prime-power Bragg peaks in the zeros' structure factor [17, 27]) supply the "crystal" half as a single spectrum but not the superposition model; Morán Ledezma [19] treats the explicit formula as Bohr-space spectral measures (a value-distribution statement, not a zero-statistics one).

8.3 Contrast with the de Bruijn–Newman heat flow

Our deformation should not be confused with the de Bruijn–Newman heat flow, though the two sides of the bridge stand in different relations to it. In the heat-flow picture the family $H_t = \mathrm{e}^{t\Delta} H_0$ is, for *every* value of the flow parameter t, an entire function of order one carrying the full infinite complement of ξ-type zeros at Weyl density; the backward heat flow merely transports these already-existing zeros (by a Calogero–Moser-type ODE) onto or off the real axis, and the de Bruijn–Newman constant Λ (now known to satisfy $0 \le \Lambda \le 0.22$, with RH $\iff \Lambda = 0$ after 22, 25) marks the threshold at which an existing configuration has become entirely real. No zeros are created or destroyed; the question is purely *when* a fixed-cardinality configuration becomes all-real.

The ζ side: creation, not real-ification. On the ζ side the contrast is sharp. The harmonic-count knob K interpolates from a genuinely *zeroless* endpoint — the bare continuous integral (1), equivalently the $K = 0$ warp object, which has no non-trivial zeros at all — to the full Dirichlet series, and the non-trivial zeros are *born* as K leaves 0 and only then migrate to the critical line (Observation 2). There is no fixed pre-existing configuration being rearranged, and no Λ-type threshold: this is creation-and-migration from a zeroless endpoint, not the real-ification of a pre-existing infinite zero set.

The η side: the nearer cousin, but still distinct. On the η side the analogy is closer, and we flag it rather than paper over it. The $K = 0$ ground string there is *not* zeroless: it is $-F(s) + \frac{1}{2}$, which already carries an infinite zero string at the full Riemann–von Mangoldt density (Observation 1); as $K \to \infty$ those zeros are *transported*, not created — the same fixed-cardinality, density-preserving motion of a pre-existing infinite zero set that the heat-flow picture has. Three features nonetheless keep it distinct. First, the destination is a pair of *vertical* critical lines ($\sigma = \frac{1}{2}$ and $\sigma = 1$), not the real axis of a symmetrised ξ. Second, the motion is a density-conserving *split*: the single ground string is *sorted* onto two lines with the exact bookkeeping of (7), a bifurcation driven by the arithmetic $1-2^{1-s}$ comb for which the heat flow — which keeps one configuration on one axis — has no analogue. Third, the deformation parameter is a harmonic count, the dynamics are not a Calogero–Moser gradient flow, and there is no value of K at which an all-line configuration is first achieved — the lines are reached only in the limit. So the bridge separates a genuine "born from nothing" mechanism (ζ) from a "transported and sorted" one (η); only the latter resembles the heat flow, and even then only in the weak sense that a pre-existing infinite zero set is moved by a parameter.

8.4 Contrast with Taylor-section and resummation routes

Our rate is algebraic, $O(1/K)$ (13). This distinguishes the construction from the Taylor sections of ξ, whose zeros converge *super-exponentially* [14], and from the Borel/resurgent resummation of the *entire* Euler–Maclaurin remainder [6], which retains no truncation parameter and births no zeros. The shared sawtooth tail (13) is what makes *our* family uniformly $O(1/K)$ across all routes.

9 Reproducibility

Every figure in this paper is produced by a driver of the same name in the accompanying repository, and every driver self-validates its identities to full precision before it plots, so each figure doubles as a numerical certificate. The eleven figures are regenerated together by a single command:

`python repro.py` (regenerates all figures); `pytest` (fast self-checks; `pytest -m slow` for the high-precision suite).

The drivers are pure `mpmath` except `eta_two_component.py`, which additionally reads a cached table of Riemann zeros. The code targets Python 3.9 and is released under the MIT license. Full setup instructions, a figure-by-figure walkthrough (`RESULTS.md`), and the annotated bibliography accompany the source at `https://github.com/bradleypmartin/dirichlet-bridge`.

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